

# Statistical methods in bioinformatics

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## 3.1 Introduction

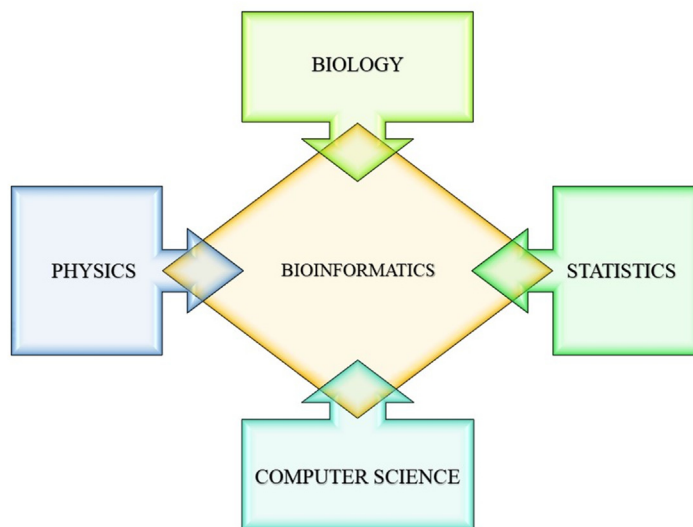
Statistics is the study of collecting, mining, organizing, analyzing, and presenting data in a manner that draws a meaningful conclusion about the set of data. This allows us to utilize smaller sets of samples for making precise and wise interpretations about a larger population. When statistics are incorporated into bioinformatics, it aids in an easier and much faster analysis of biological data. The main objective of statistics and bioinformatics is to assist researchers in acquiring equitable solutions to various biological problems using computational data analysis. Statisticians bring not only a distinct viewpoint but also a set of skills to this approach, putting them at the center of computational research. Understanding fluctuations and uncertainties in quantification is among the fundamental characteristics that distinguish a statistician from other mathematicians. These are critical factors to take into consideration when developing reliable techniques for biological discoveries and verification, particularly when dealing with complex, high-dimensional data like that found in genomic information. The decisions made when designing samples have big effects on the analysis that comes after. Decision making can influence the multistep processing techniques while reducing data loss caused by extraction methods and may be responsible for propagating the errors in the entire process, so accurate decision making is prioritized using the standard protocols to reduce the errors. In the next few sections, we will discuss about some statistical methods used to make sense of biological data.

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## 3.2 Statistics at the interface of bioinformatics

The word “bioinformatics” refers to a multidisciplinary area in which computational researchers, mathematical data analysts, systems biologists, and statisticians investigate multiple dimensions of biological data, including its storage, retrieval, organization, and detailed evaluation. Due to the numerous obstacles presented by this complicated discipline, bioinformatics must be multidisciplinary. It is not possible for a single researcher to possess the medical, biological, computational, data repository management, mathematical analysis, and statistical skills and knowledge

necessary to locate and validate the large amount of scientific information contained in the outputs of these technologies. In bioinformatics, some of the most important things to study are sequence alignment, gene discovery, gene assembly, protein structure alignment and its predictions, gene expression estimation, protein-protein interaction, and evolutionary modeling. Since the bulk of biological information is not stored in a database reduced to a single, flat record, data extraction from structured information is especially significant for bioinformatics application domains. Bioinformatics databases are set up with elements that are connected to each other and to other elements by links that describe a complex structure. The unique viewpoint and skill set of statisticians position them at the core of this process. Awareness of variability and unpredictability in assessment is one of the major characteristics that set statisticians apart from all other quantitative disciplines. These are critical factors for constructing reproducible techniques for biomedical exploration and testing, particularly when dealing with complex, high-dimensional data such as those observed in genomics. Statisticians are “data scientists” who know the affects of sampling design decisions on later stage analysis, how biases can spread through multi-step processing methods, and how information can be lost if feature extraction methods are too simple. They are experts in inferential reasoning, which enabled them to recognize the significance of multiple statistical revisions to avoid reporting spurious results as discoveries and to structure algorithms to hunt high-dimensional spaces and construct predictive statistical models while obtaining precise measurements of accuracy (Fig. 3.1).



**FIG. 3.1**

Bioinformatics, an interdisciplinary field of science, aims at using concepts from major branches like statistics, biology, computer science and physics.

There are two types of statistical methods used in analyzing biological data: first one is descriptive statistics while the other one is inferential statistics.

For describing data from a representative cohort, descriptive statistics utilize the mean and standard deviation. It is a type of statistical analysis that looks at a population using math, graphics, and tables. Descriptive statistics assist in portraying data in a more meaningful fashion, allowing for easier data interpretation.

Inferential statistics views information as a subgroup of a given sample, using datasets from that sample to draw conclusions and make calculations about that group. Inferential statistics allows one to establish generalizations regarding a group using sample data. Hypothesis testing, confidence intervals, and linear as well as multiple regression analysis are the most frequent inferential statistical approaches.

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### 3.3 Measures of central tendency

Before going on to statistical analysis, it is required to review the raw data. Then, the two most significant sample statistics that can be derived from a dataset are a measure of the sample distribution's central tendency and the data's dispersion around this central tendency. These descriptive statistics are necessary for inferential statistical analysis. In simple terms, "central tendency" is a technique for describing the dataset's core. It is also known as a distribution center or site. Various measures of central tendency strive to define what is variably referred to as the typical, normal, expected, or mean value of a dataset. The mean, the mode, and the median are widely utilized for the majority of data types. The mean is one of the most common measures of central tendency, and most people think it's the best. However, in some cases, either the median or the mode is preferable. Whenever there are some extreme values in the given data, the median seems to be the recommended measure for the central tendency ([Smucker et al., 2018](#)).

#### 3.3.1 Mean

Statistically, a dataset's mean is a single number that indicates the midpoint or average value of all the data points. One way to gauge central tendency is through the use of the mean, which is sometimes known as the arithmetic average. For determining the mean, the sum of all data elements is multiplied by the number of times it has been observed. This is the most common and widely used statistic. It can be used to characterize a population sample containing a singular value representing the data's center.

$$\text{Mean} = \frac{\text{Sum of all the observations}}{\text{Total Number of observations}}$$

Ideally, the mean represents the region in which the majority of the values in a distribution fall. The mean does not, however, necessarily locate the center of the dataset. It is susceptible to extreme data and can be skewed. This issue emerges

when outliers have a considerable influence on the mean. As the skewness of the distribution increases, the average moves further from the center. In some instances, the mean might be deceptive since it may not be close to the most common values. Therefore, it is optimal to use the mean to identify the central tendency in a symmetric distribution. For skewed distributions, it is frequently preferable to utilize the median, which locates the center using a different approach. The mean doesn't tell us anything about how different a distribution is, so we need to look at the standard deviation to figure out how different it is.

### 3.3.2 Median

It's a straightforward metric for determining central tendency. For information categorized at an ordinal level, that is the most appropriate indicator of average. The median in statistics is the value that divides a ranked group of data entries in half and corresponds to the 50th percentile of the dataset. It is exactly in the center of the dataset, with half the numbers below and half above. Locating the median is simple. Sort the values in the dataset from smallest to greatest. Then, find the value with the same number of data entries above and below it. The middle value is determined differently depending on whether your data has an even or odd number of values. This section shows you how to figure out the median for data that is grouped or ungrouped.

#### 3.3.2.1 Median for the grouped data

The following formula is used to find the median when the data points are in odd number:

$$\text{Median} = \left( \frac{N+1}{2} \right)^{\text{th}} \text{ term}$$

The following is the formula for calculating the median for even number of data points:

$$\text{Median} = \frac{\frac{N^{\text{th}}}{2} \text{ term} + \left( \frac{N}{2} + 1 \right)^{\text{th}} \text{ term}}{2}$$

Here, “N” represents the entire number of elements/observations in the sample dataset and “th” indicates the (N)th number.

#### 3.3.2.2 Median for the ungrouped data

If the data is well grouped then we use the following formula to find the median:

$$\text{Median} = l + \left( \frac{\frac{N}{2} - cf}{f} \right) \times h$$

here, “ $l$ ” is the median class lower limit, “ $cf$ ” represents the cumulative frequency of class previous to the dataset’s median class. “ $f$ ” signifies the frequency of median class and “ $h$ ” represents the size of class.

One can gain an impression of a dataset’s dispersal by contrasting its median with the mean. The set of data is almost uniformly dispersed across lowest and highest values whenever the mean and median value are identical. When the sample distribution is reported to be skewed, the median is a more accurate estimate of central tendency than the mean since it is less vulnerable to outliers. Extreme values drive the mean distant from the distribution’s center, rendering it potentially deceptive.

### 3.3.3 Mode

Similar to the mean and median, the mode is a statistical method to calculate the central tendency. The mode is by far the most prevalent value in a data set, i.e., the characteristic or value that occurs most frequently in any particular dataset. To put it another way, the mode of a set of numerical data is the variable occurring with the greatest frequency in that dataset. It is an averaged metric that can be applied to nominal variables. There may be more than one mode in a dataset, which is different from other measures of central tendency. A data set can be bimodal (having two modes), trimodal (having three modes), multimodal (having  $>3$  modes), or no-model (having no modes). One of its best features is that it can be used for any kind of data, unlike the mean and the median, which can’t be determined for nominal data. In the case of normally distributed data, its mode has the same numeric values as observed in the mean and median, although it can be quite distinct in severely skewed patterns. Mode is unaffected by the most and least extreme numbers in quantitative datasets. So, it can give insights into almost any set of data values, no matter how the data is organized. However, the statistical metric is not without its own limits. For example, it cannot be dealt with mathematically any further. Therefore, it cannot be utilized for a more comprehensive study. In addition, because mode is not based on all values in the dataset, it is challenging to draw inferences about the dataset using mode alone (Krzywinski and Altman, 2014a).

### 3.3.4 Percentiles, quartiles and interquartile range

Percentiles are remarkably versatile and can be used to produce a relative ranking, to divide a dataset into equal proportions, to determine a distribution’s central tendency, and to assess its dispersion. Also, quartiles are a means of describing the distribution of data. Every collection of “ $X$ ” intervals that create boxes having an identical number of observations is referred to as a “quantile” of that variable “ $X$ ”. The audience is probably familiar with percentiles, which are boxes comprising 1% of information. Another popular method is the quartile, which uses “ $4X$ ” intervals for defining boxes holding 25% of the information. Whenever distributions contain skewness or excessive kurtosis, the interquartile range (IQR) is usually used instead of the standard deviation since the standard deviation cannot offer

dependably decipherable information regarding non-Gaussian distributions. The median represents the 50th percentile. This value halves a given dataset. Half of the results fall under the 50th percentile, while the other half exceed it. Based on percentiles, quartiles are the numbers that split your data into quarters. The 25th percentile corresponds to the first quartile, commonly known as Q1 or the lower quartile. One-fourth of the scores fall below this threshold, while three-quarters exceed it. The median, often known as the second quartile or Q2, is just the value of the 50th percentile. Half of the scores are significantly higher than average, while the other half are significantly lower. The 75th percentile corresponds to the third quartile, commonly known as Q3 or the upper quartile. One-quarter of the scores exceed this number, while the remaining three-quarters are below. In descriptive analysis, the interquartile range is often referred to as the “mid-spread,” the “mid 50%,” or the “H-spread.” It is a statistical dispersion indicator dependent on the division of a data frame across quartiles. The IQR is defined as the gap between the 75th and 25th percentiles, establishing a region of “X” that encompasses the central 50% of distributions, but is not always centered around the average as well as the median (Fig. 3.2) (Krzywinski and Altman, 2014b).

$$IQR = Q3 - Q1$$

The q-q plot, also known as the “quantile-quantile plot,” is a method for determining whether two sets of data originate from the same population with the same distribution (Fig. 3.2). By plotting the quantiles of two probability distributions

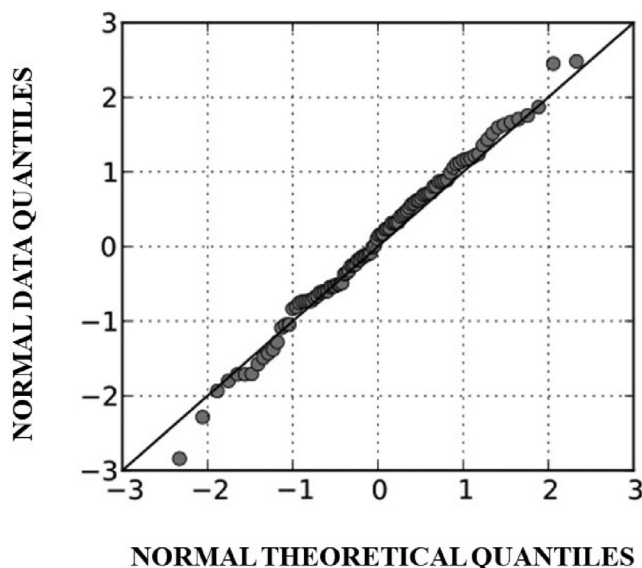


FIG. 3.2

A normal quantile-quantile plot (QQ-Plot).

against each other, Q-Q (quantile-quantile) plots are a very important way to compare and analyze them visually. If the two population distributions we are trying to compare are exactly the same, then the observations on the Q-Q plot will flawlessly continue lying on a straight line following “ $y = x$ .” Q-Q plots are utilized to compare the distributional shapes of two groups. They illustrate how placement, size, and skewness are similar or dissimilar between the two distributions. Q-Q plots can be used to compare data sets or models of possible outcomes.

### 3.4 Skewness and kurtosis

When sample data considerably deviates from the norm, skewness quantifies the ensuing imbalance. Sometimes, the normal distribution appears to be skewed to one side. This is because the possibility of data being greater than or less than the mean is greater, resulting in an asymmetrical distribution (Fig. 3.3) (Royston, 1992).

As we know, a normal distribution takes the form of a bell curve, any deviation from this normal bell curve leads to skewness, suggesting that the data are not equally distributed. The asymmetry may take two forms:

1. **Positively Skewed:** In a distribution with a positive skew, the values are more packed on the right side, while the left tail is more dispersed. Consequently, the statistical results are left-skewed. Therefore, the mean, the median, and the

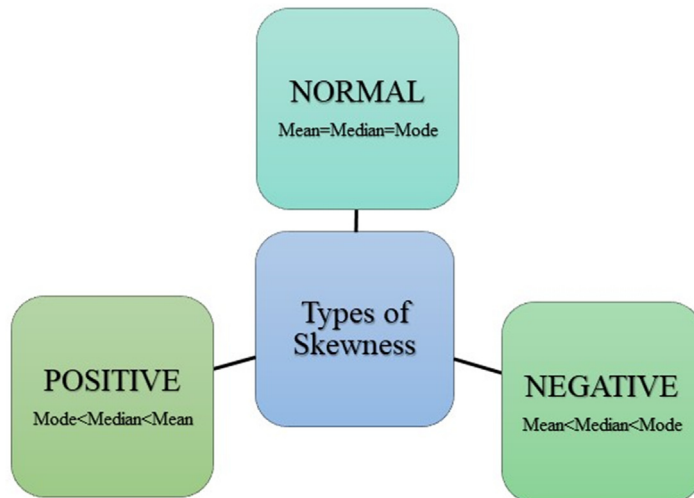


FIG. 3.3

Three types of skewness: Normal where mean, median and mode, all are equal; Positive where mean is greater than median and mode; Negative where mode is greater than mean and median.

mode are always positive. In this type of distribution, the mean is bigger than the median and the mode.

- 2. Negatively Skewed:** The right side of the distribution has a higher density of sample points when it is negatively skewed. The average, median, and mode shift to the right as a result. Therefore, these numbers are always negative. In this distribution, the mode exceeds the median while falling short of the mean (Fig. 3.4).

Kurtosis, similar to skewness and is applied to discover data sample outliers. It provides the proportion of outliers present in total. The data may have heavier tails and a flatter peak, comparable to when a distribution is punched or squashed. It is referred to as negative kurtosis (polykurtic). Positive kurtosis indicates a distribution in which the top curve is steeper, indicating that the sample distribution is being pulled upwards (leptokurtic). A normal kurtosis (mesokurtosis) indicates a normal distribution (Fig. 3.5).

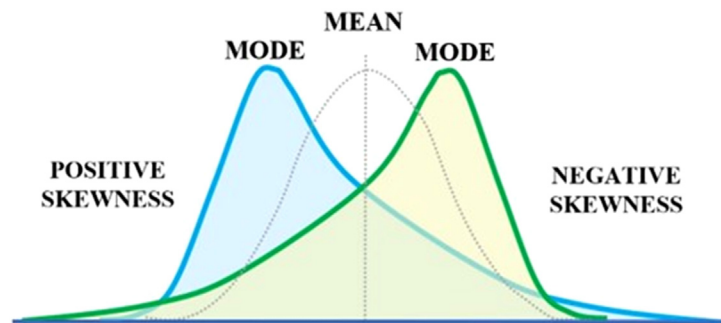
The expected value of kurtosis is 3. There is symmetry in this distribution. Positive kurtosis is present if the kurtosis value is greater than three. In this instance, the range of the kurtosis value is between 1 and infinity. A kurtosis of fewer than three also denotes a negative kurtosis. A negative kurtosis has a value range of 2 to infinity. The peak gets higher as the kurtosis value increases (Fig. 3.6).

The typical distributions of variables in everyday life are seldom flawless. Both the skewness and kurtosis coefficients show how much a dispersion is different from data that is normally distributed. Thus, we may say that skewness and kurtosis are used to define the width and height of the normal distribution. The data's horizontal drag is represented by skewness. Kurtosis is used to calculate the vertical drag or the height of the peak, as it reflects how dispersed the data is.

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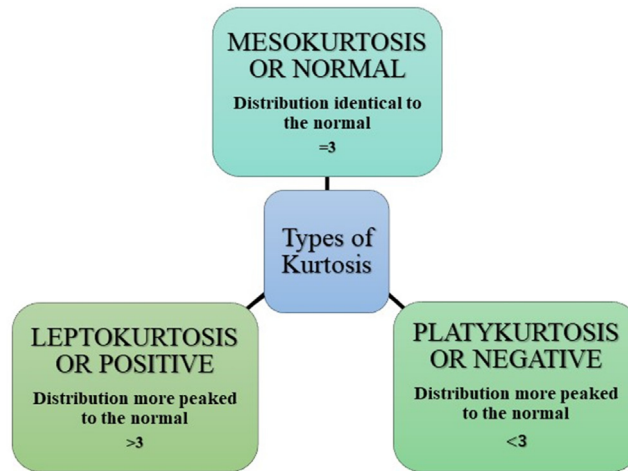
### 3.5 Variability and its measures

Variability is a statistical method that usually represents the dispersion within a given dataset. A measure of central tendency provides the mean, whereas a measure



**FIG. 3.4**

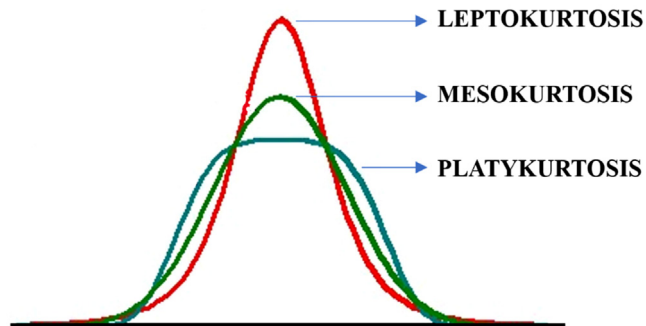
A schematic diagram showing positively skewed and negatively skewed distribution.

**FIG. 3.5**

Three types of kurtoses: Normal where distribution is identical to normal; Positive where distribution is more peaked toward the normal; Negative where distribution is less peaked toward the normal.

of variability specifies the proportion of additional data points that are likely to deviate from the mean. Within the context of a value distribution, variability is taken into account. A low dispersion suggests that the sampled data points are firmly grouped around the distribution's center. The greater the dispersion, the further apart they are likely to fall (Hazra and Gogtay, 2016).

Analysts usually use the mean to talk about the center of a group or even a whole system. Since the mean is important, individuals frequently react more to variability. When the variability of a distribution is minimal, the values in a dataset seem to be more stable. However, whenever the variability is increased, the data points become

**FIG. 3.6**

A schematic diagram of different types of kurtoses representing their peaks.

more different and the likelihood of extreme results increases. Therefore, understanding variability enables you to comprehend the chance of odd occurrences. In the sections that follow, we will give an in-depth look at some of the measures of variability used to understand biological data.

### 3.5.1 Variance

The prediction of a randomized variable's squared deviation from its mean is called its variance. It determines how distant a group of figures or information are from its mean. The variance compares each value to the mean in order to include all values in the computation. A huge variance shows that the sample units are far off from its mean and each other. To generate this statistic, the squared discrepancies between the data points and the mean are computed, added, and then divided by the total number of observations in the sample dataset. There are two formulas to compute the variance of the sample, depending on whether we are determining the variance for the whole population or using a subset to predict the sample variance of the entire population (Dakhale et al., 2012).

To assess the variation of the whole population, following formula of *Population Variance* is used:

$$\sigma^2 = \frac{\sum (X - \mu)^2}{N}$$

here, “ $\sigma^2$ ” represents the variance's population parameter, “ $\mu$ ” represents the population parameter for the population mean, and “ $N$ ” denotes the number of data points that should represent the overall population.

While, to find the variance of the population using a small subset of population, following formula of *Sample Variance* is used:

$$s^2 = \frac{\sum (X - M)^2}{N - 1}$$

where, “ $s^2$ ” is the variance for the sample taken from the population and “ $M$ ” denotes the sample's mean. The inclusion of “ $N - 1$ ” in the denominator compensates for the propensity of samples to underestimate the population's variance.

### 3.5.2 Standard deviation

The standard deviation shows how far each data point is from the population mean on average. It goes down when the values in a dataset are closer together, indicating a lower dispersion of data values. On the other hand, when there are more differences between the values, the standard deviation goes up because the standard deviation goes up, signifying a higher dispersion in the data values. The standard deviation uses the data's original units, which makes it easier to understand. So, the standard deviation is the most common way to measure how much something can change within a population as well as for a specific subset of that population. The standard

deviation is determined by taking the square root of the sample variance, which is a metric that represents the dispersion or variability of a sample in relation to its mean, which represents how far a set's measurements deviate from its average/expected value.

To find the variance of the whole population, following formula of *Population Standard deviation* is used:

$$\sigma = \sqrt{\frac{\sum (X - \mu)^2}{N}}$$

here, “ $\sigma$ ” stands for the population standard deviation. “ $X$ ” signifies the values present in the data distribution, “ $\mu$ ” signifies the population mean while “ $N$ ” stands for the total number of observations in the population sample.

To find the variance of a sample drawn out of a population, following formula of *Sample Standard deviation* is used:

$$s = \sqrt{\frac{\sum (X - \bar{X})^2}{N - 1}}$$

where, “ $s$ ” signifies the standard deviation for the sample drawn out of a particular population and “ $\bar{X}$ ” stands for the mean of the sample.

Remember that the square units are used to measure variance, and as a result, the square root helps to restore the value to its native units. Standard deviation is used as a parameter for the whole population and is shown by the symbol “ $\sigma$ ”, while as an estimate from a sample, however, it is shown by the symbol “ $s$ ”. To figure out the standard deviation, figure out the variance as shown above, and then take the square root of the result. The standard deviation is a number that is like the mean absolute deviation and is considered the most accurate indicator of variation. Both use the original units of the data and compare the data values to the mean to find out how much they vary. But there are differences.

### 3.5.3 Standard error

Standard error, sometimes known as “standard error of the mean,” is an inferential measure that depicts, in simple terms, the degree to which the data from a sample are representative of the total population. For instance, if you conduct a survey of Delhi residents, you will acquire a sample of data that is representative of a portion of Delhi residents. Different types of the same population might thus have different results, so it is essential to evaluate the significance of your findings. The standard error represents the distance between two means when comparing the mean of the sample data to the mean of the complete population on a distribution. How much would the sample mean change if the same study was conducted on a different group of Delhi, India residents? To figure out the standard error, divide the standard deviation by the square root of the sample size “ $N$ ” for the whole population.

$$SE = \frac{\sigma}{\sqrt{N}}$$

In statistics, sample information is utilized to comprehend larger populations. Probability sampling, in which pieces of a sample are picked at random, enables the collection of data that is likely to be representative of the population. However, there will be some sampling error with probability samples. In terms of measurements like means and standard deviations, a sample can never exactly represent the population from which it was drawn. By calculating the standard error, it is feasible to assess the sample's resemblance to the population and derive the relevant conclusions. A huge standard error infers that the mean of the sample population is not equitably spread around the population mean, suggesting that this sample may not be successfully representing the population. A negligible standard error shows that the mean of the sample is clustered closely around the population mean, recommending that the sample effectively represents the population. By increasing sample size, the standard error can be decreased. Using a huge, random sample is the most effective method for reducing sampling bias.

What distinguishes standard deviation from standard error?

The standard deviation is a marker of to what extent a collection of observations deviates from its mean or predicted values, whereas the standard error is basically an estimation of how widely the collection's average deviates from the genuine population average. The standard deviation is invariably greater compared to the standard error. Let's compare the distinctions between them. The most significant distinctions are: Standard error reflects variability across numerous samples of a population, whereas standard deviation depicts variability within a specific subset of that population. The standard error is a statistical descriptive metric that can only be approximated, whereas the standard deviation is an inferential statistic that can be derived from sample data. The formula for standard error is the standard deviation divided by the square root of the sample size. The formula for standard deviation is the square root of the sample variance.

### 3.5.4 Coefficient of variation

The coefficient of variation (CV) is a statistical metric of relative variability that represents the amount of standard deviation relative to the population mean. It is a common, non-united measurement that tries to determine the variability of dissimilar groups and attributes. It's also referred to as the "relative standard deviation" (RSD). It comes in particularly handy when evaluating the level of variance between two datasets, since the allowable limit inside the variance grows as the standard deviation number rises and the information becomes less exact. Utilizing a simple ratio, the coefficient of variation is calculated. The variance can be estimated by dividing the standard deviation by the sample mean.

$$CV = \frac{\text{Standard deviation}}{\text{Mean}}$$

For instance, the delivery time of a courier service is measured in minutes. The average delivery time is 30 min, with a standard variation of 6 min, and hence, 0.20 is the coefficient of variation for the parcel delivery case. This value represents the magnitude of the standard deviation relative to the mean. Analysts typically represent the coefficient of variation as a percentage, so for this situation of parcel delivery, the standard deviation is 20% of the mean size. If the value is one or 100%, then the standard deviation and the mean are equal. Less than one implies that the standard deviation is less than the mean, while greater than one indicates that it is greater than the mean. In general, bigger values correspond to a higher degree of relative variability.

#### ***3.5.4.1 Comprehending the source of variability for analysis***

Every statistical study presumes that whatever is seen in the samples could be extrapolated to the entire populace. We might be fairly sure if our findings are generally applicable to a community under investigation if we recognize the samples. We can cope with several causes of variation, including biological as well as technological ones, which, if disregarded, can have a significant effect on the statistical analysis as well as the results. Scientists doing omics studies are familiar with a variety of origins of unpredictability. Details can comprise information regarding dates and manner of data gathering or processing, the procedure followed, the scientist doing preliminary studies, computational tools, and a variety of other discovery details. We can adjust for such possible disparities in the data using both quantitative and technological methods, but we'll require such evidence first. We could set up the study so that all the data is collected in the same way by the same scientist, or we could use controls to make up for these differences in the statistical analyses. Determinants of diversity in the research group are also crucial. These could be unfamiliar to individuals conducting omics research, yet they certainly could have an impact on overall results. When making a prediction based on different factors, like the effect of the drug, it is important that the samples that got the drug and the ones that didn't get it are as similar as possible. Whenever the distinguishing factor is an inherent contradiction across data, such as when comparing the expression of genes in people with the presence or absence of an illness, it gets increasingly problematic. It's hard to locate infected and uninfected people that are otherwise identical, so any changes in expression could be attributable to anything besides the illness.

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### **3.6 Different types of distributions and their significance**

A "distribution" is basically a set of data or scores associated with a variable. Typically, these values are sorted from lowest to highest and can then be shown graphically. The most popular distribution is the Gaussian distribution, commonly known as the "normal" distribution. This distribution provides a standardized mathematical formula that may be used to calculate the probability for each sample observation. The probability density function, which indicates the grouping or density of the data,

is represented by this distribution. The likelihood that an observation has a value equal to or less than a specified value is also calculable. A cumulative density function offers an overview of various data relationships. In practice, we might think of a distribution as a statistic that shows the relationship between observations in a data point. For example, we may be interested in people's height, with individual heights representing domain observations and heights 140–180 cm being the sample space. The distribution is a mathematical function that depicts the relationship between different height readings. Many data points are congruent with well-known and acknowledged mathematical functions, such as the Gaussian distribution. Modifying the function's parameters, such as the mean and the standard deviation in the case of a Gaussian distribution, enables the function to be tailored to the given data. When a distribution function is known, it may be utilized as a shortcut for defining and computing related quantities, such as observation probabilities and domain connections (Islam and Al-Shiha, 2018).

### 3.6.1 Probability distributions

The probability distributions are statistical expressions that represent all of the potential readings and probabilities for a random vector inside a particular region. This region would be limited by the lowest and greatest input states, and wherever that potential quantity would be displayed just on the probability density function, it would be determined by a variety of parameters. The average, standard deviation, skewness, and kurtosis of distributions are among such parameters. The normal distribution, also known as the “bell-shaped curve,” represents among the most frequent probability distributions; however, there are many more. Usually, this method of obtaining information about a phenomenon determines its probabilistic model. The probability density is the name for such a procedure. In the data design phase, there are two types of probability distributions that are used for different reasons: the discrete probability function, and the second is the continuous probability function (Khakshooy and Chiappelli, 2018).

Some examples of these classes include the normal distribution, chi-square test distribution, binomial distribution, and Poisson distribution. Distinct probability distributions describe distinct information generating methods as well as fulfilling various purposes. The binomial distribution calculates the estimated likelihood of the occurrence of events multiple times across a set of experiments, knowing the overall likelihood of an incident within every attempt. Because just 1 or 0 is a legitimate answer, a binomial distribution is discontinuous rather than being a continuum.

### 3.6.2 Continuous probability function

#### 3.6.2.1 Normal distribution

The normal distribution is a homogeneous/symmetric probability distribution about the mean that shows data closer to the mean are more common than data farther from

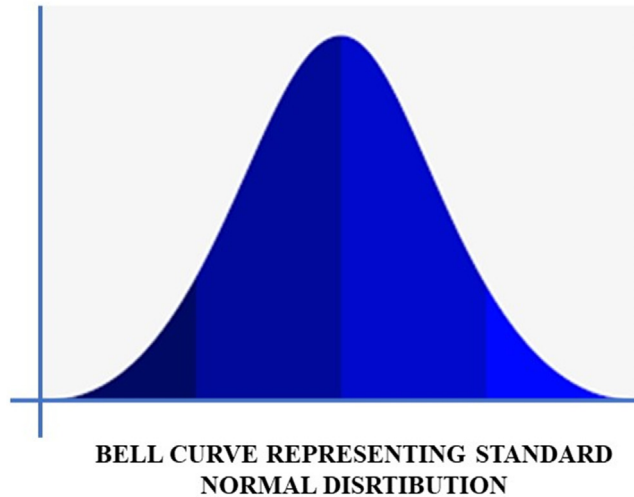
it. A bell curve (Fig. 3.7) appears on a diagram representing a normal distribution. A random variable with a normal sampling distribution (also called a Gaussian distribution) is called a normal deviation. Although all normal distributions are symmetric, not all symmetric distributions are normal. The mean and standard deviation fully characterize the normal distribution, showing that the distribution is unbiased but has kurtosis. The mean (average) equals zero and the standard deviation equals one, which define a normal distribution with zero skewness and 3 kurtoses (Fig. 3.7).

In a normal distribution, approximately 68% of the collected data falls within  $\pm 1$  standard deviation of the mean, 95% of the data falls within  $\pm 2$  standard deviations of the mean, and 99.7% of the data falls within  $\pm 3$  standard deviations of the mean. As a clear contrast to binomial distribution, the normal distribution is continuous in nature, meaning that all possible values are displayed (as contrasted to 0 and 1 with no intermediate value).

When, “ $\mu = 0$ ” and “ $\sigma = 1$ ”, it is called as Probability density function denoted by “ $f(x)$ ”, “ $\mu$ ” refers to mean (median and mode also) of the distribution and “ $\sigma$ ” refers to standard deviation.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Normal distribution integrals are the Cumulative density function of conventional normal distributions, as depicted in equation below. This is typically represented with a capital Greek letter “ $\Phi$ ”.



**FIG. 3.7**

A bell curve illustrating the conventional normal distribution.

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}} dt$$

### 3.6.2.2 Continuous uniform distribution

A continuous uniform distribution is a symmetric likelihood distribution, commonly known as a “rectangular distribution” as it forms a rectangle on a graph. This type of continuous distribution is a method for describing an experiment whose outcomes are random and fall within a given range. Minimum and maximum values are specified via the parameters “a” and “b”, which set the limits. The interval range for this distribution can be closed, as in  $[a, b]$ , or open, as in  $(a, b)$ . Thus, the distribution is commonly represented as  $U(a, b)$ , where “U” stands for “uniform distribution.” The gap between the interval limits determines the length of the interval. All intervals of the same length are possible on the distribution’s support. It is the probability distribution with the highest entropy for a random variable “x” that is not limited in any way other than being part of the distribution.

In relation to a continuous uniform distribution, the probability density function is represented by:

$$f(x) = f(x) = \begin{cases} \frac{1}{b-a}, & \text{For } a \leq x \leq b, \\ 0, & \text{For } x < a \text{ or } x > b \end{cases}$$

The value of “ $f(x)$ ” at the two limits a and b are often insignificant since they do not affect the integrals of “ $f(x)dx$ ” across any interval. Occasionally they are selected as zero, and sometimes they are selected as “ $\frac{1}{b-a}$ ”. In the situation of estimating using the approach of maximum likelihood, the latter is suitable.

$$F(x) = \begin{cases} 0, & \text{For } x < a \\ \frac{x-a}{b-a}, & \text{For } a \leq x \leq b \\ 1, & \text{For } x > b \end{cases}$$

Cumulative distribution function is represented above for continuous uniform distribution. The cumulative distribution function is expressed in the notation of the mean and the variance as:

$$f(x) = \begin{cases} 0, & \text{For } x - \mu < -\sigma\sqrt{3} \\ \frac{1}{2} \left( \frac{x-\mu}{\sigma\sqrt{3}} + 1 \right), & \text{For } -\sigma\sqrt{3} \leq x - \mu < \sigma\sqrt{3} \\ 1, & \text{For } x - \mu \geq \sigma\sqrt{3} \end{cases}$$

### 3.6.2.3 Log-normal distribution

A probability distribution with a normally distributed logarithm is called a log-normal distribution, or Galton distribution. A random variable is lognormally oriented if its logarithm follows the same pattern as the normal distribution. This

type of distribution often has skewed distributions with minimal mean values, massive variance, and all positive values.  $\text{Log}(x)$  only works when  $x$  is positive, so values have to be positive. The mean “ $\mu$ ” and the standard deviation “ $\sigma$ ” tell us what the probability density function for Log-normal distribution is.

A random variable “ $X$ ” that is positive is said to have a log-normal distribution if its natural logarithm has a normal distribution with the mean and the variance.

$$\ln(X) \sim \mathcal{N}(\mu, \sigma^2)$$

If “ $\varphi$ ” represent the probability density function for the distribution  $N(0,1)$ , then it follows that:

$$f_X(x) = \varphi\left(\frac{\ln x - \mu}{\sigma}\right) \frac{1}{\sigma x}$$

And if “ $\Phi$ ” represent the cumulative distribution function for the distribution  $N(0,1)$ , then it follows the equation:

$$F_X(x) = \Phi\left(\frac{(\ln x) - \mu}{\sigma}\right)$$

#### 3.6.2.4 Exponential distribution

The time between events in a Poisson point process, where events occur repeatedly, autonomously, and at a steady average rate, has an exponential probability distribution, and this type of distribution is known as an exponential distribution. This is a special example of the gamma distribution. It’s the continuous version of the geometric distribution, and the most important thing about it is that it doesn’t remember anything. It is used for more than just studying Poisson point processes.

For an Exponential distribution, Probability distribution function can be defined as:

$$f(x; \lambda) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Here,  $\lambda > 0$  is the exponential distribution variable, which is also termed as the rate parameter. The valid range for this type of distribution is between [0 and infinity). For a random variable “ $X$ ” having an exponential distribution, then we will write  $X \sim \text{Exp}(\lambda)$  and the exponential distribution can be divided into infinite number of parts.

Cumulative distribution function can be defined as below for an exponential function.

$$F(x; \lambda) = \begin{cases} 1 - e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

### 3.6.3 Discrete probability function

A discrete distribution reflects the likelihood of occurrence for each discrete random variable value. For example, let's take the example of rotten eggs in a tray. The number of total rotten eggs out of a total number of eggs will be the discrete probability function. In a discrete probability distribution, each possible value of a random variable is associated with a probability that is positive and greater than zero. Some of the significant probability distribution functions are discussed below.

#### 3.6.3.1 Binomial distribution

A binomial distribution could be conceived of as the likelihood of a successful or unsuccessful output in a multi-step investigation or assessment. The binomial distribution is a type of probability distribution that takes into account two different possible outcomes. It's a popular approach to evaluating probability distributions that is utilized in a lot of statistical methods.

In instance, we use  $X \sim B(n, p)$  if the random variable “ $X$ ” approaches the binomial distribution for parameters  $n \in \mathbb{N}$  and  $p \in [0, 1]$ . The probability mass function tells us how likely it is that we will get exactly “ $k$ ” successes out of “ $n$ ” separate Bernoulli trials:

$$f(k, n, p) = \Pr(k; n, p) = \Pr(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

for  $k = 0, 1, 2, \dots, n$ , where  $\binom{n}{k} = \frac{n!}{k!(n-k)!}$  is the binomial coefficient, that's why this distribution is called the binomial distribution. This binomial distribution formula can be explained like this: “ $k$ ” successes happen with a chance of “ $p^k$ ”, and “ $n - k$ ” failures happen with a chance of “ $(1 - p)^{n-k}$ ”. The “ $k$ ” successes, on the other hand, can happen anywhere in the “ $n$ ” trials, and there are “ $\binom{n}{k}$ ” different ways to spread them out over the “ $n$ ” trials.

One possible way to express the cumulative distribution function is as follows:

$$F(k; n, p) = \Pr(X \leq k) = \sum_{i=0}^{\lfloor k \rfloor} \binom{n}{i} p^i (1 - p)^{n-i}$$

Where, “ $\lfloor k \rfloor$ ” is the “floor” under “ $k$ ”, i.e., the highest integer less than or equal to “ $k$ ”.

#### 3.6.3.2 Bernoulli's distribution

The Bernoulli distribution is a discontinuous likelihood distribution. With probability “ $p$ ” the Bernoulli distributed random variable has the value 1, but with

probability “ $q = 1-p$ ”, it has a value of zero. In a less formal context, it can be regarded as a statistical model for the assortment of probable consequences of any individual testing that poses a true-or-false questions. These questions result in boolean-valued responses, where a value is considered “success” or “yes” with a likelihood of “ $p$ ” and “failure” or “no” with a likelihood of “ $q$ ”. It can be used to epitomize a coin toss, with 1 and 0 signifying “heads” and “tails” and “ $p$ ” representing the probability that the coin will drop on “heads” or vice versa. If a coin wasn’t fair, it would have a probability of “ $p \neq 1/2$ ”.

The binomial distribution has several subtypes, including the Bernoulli distribution that only applies when only one trial is done (so for this type of binomial distribution, “ $n$ ” would be 1). Additionally, this is a specific example of the two-point distribution, in which the potential consequences do not have to be either 0 or 1.

If “ $X$ ” is a random variable and it follows this distribution, then the following holds true:

$$\Pr(X=1)=p=1-\Pr(X=0)=1-q$$

This distribution’s probability mass function “ $f$ ” across all potential outcomes  $k$  is:

$$f(k;p)=\begin{cases} p, k=1 \\ q=1-p, k=0 \end{cases}$$

### 3.6.3.3 Poisson distribution

The Poisson distribution is an example of a discontinuous likelihood distribution that is used in probability and statistics. If the rate of events is known to be constant, it illustrates how likely it is that a certain number of actions or events will arise in a definite period of time or place, regardless of how long has passed since the last occurrence. How many events occur in a given distance, area, or volume can be determined using the Poisson distribution. A poisson distribution possesses positive skewness, and whenever the average of a range is large, it resembles a normal or symmetric distribution. A Poisson distribution’s form varies; for instance, a lower average Poisson distribution shows severe skewness, featuring 0 as its mode. Having a tail spreading toward the right, every value is forced up against 0.

If “ $X$ ” a discrete random variable and has a Poisson distribution with the limitation “ $\lambda > 0$ ”, then the probability mass function for “ $X$ ” is given by the following equation:

$$f(k;\lambda)=\Pr(X=k)=\frac{\lambda^k e^{-\lambda}}{k!}$$

Where, “ $k$ ” is the number of events ( $k = 0, 1, 2 \dots n$ ), “ $e$ ” is the Euler’s number having a constant value of 2.71828 ...

### 3.6.4 Normal distribution and normal curve

A normal distribution is a type of symmetrical and balanced likelihood distribution around the mean that illustrates that data close to the mean occur more frequently than data far from the mean. On a graph with a Gaussian distribution or a normal distribution, you will observe a bell curve (discussed earlier in the section on probability distributions).

Although all normal distributions are symmetrical, not all symmetrical distributions are normal. The normal distribution's mean and standard deviation clearly describe it, demonstrating that it is not biased but rather exhibits kurtosis. As a result, the distribution is symmetrical and depicted as a bell curve. As long as the standard deviation is equal to one unity with a skew of zero and kurtosis of 3, then the distribution is considered normal.

### 3.6.5 Normal curve

With a high number of observations and a narrow class interval, a symmetrical frequency curve is produced by a histogram with the same frequency distribution of heights. This is known as the normal curve. The frequency distribution is balanced around a single peak, the mean, median, and mode will coincide in the case of a normal curve. It rises progressively from the lowest frequencies at the classification's extremes to the highest frequencies at the classification's apex. This graph is a test of SD to determine whether or not it was computed from a large, random sample representative of the universe. The form of a normal distribution or curve is highly practical and facilitates statistical analysis. It indicates the probability of occurrence by chance or the frequency with which an observation, as measured by the mean and standard deviation, can occur ordinarily in a population. By "normal," we mean normal. A peculiar finding may have a negative prognosis and also be pathological. The phrases "normal" and "abnormal" are not used in clinical situations.

### 3.6.6 Asymmetrical distribution

Distributions of various forms can be observed in nature. We have previously defined "normal distribution." It lays the groundwork for how mean and standard deviation can be used in real life to measure central tendency and variation.

Certain distributions have asymmetries or skews. Based on whether the long tail of the curve is to the left or right of the highest frequencies, they may be skewed to the left or right. Some of them are bimodal, with two peaks, as in the age-based distribution. In such situations, it is likely that two distinct groups of people are mingled together in the populace. Therefore, the sample under research is diverse.

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## 3.7 Sampling

It is not practical to enroll every member or sampling unit of the population in a scientific investigation, to examine all of the world's millions of people to determine

the prevalence of cancer, or to examine the efficacy of a medicine on every patient with a certain ailment.

In addition, a large number of investigators are necessary to execute this enormous effort, which may reduce the accuracy of a population-wide survey. Their consistency and accuracy may vary, and their acquisition will be expensive, time-consuming, and difficult. Due to these obstacles, we prefer to choose an appropriate sampling strategy. In medical research, sample data are taken from a suitably large and representative population, selected using a normal sampling method. Before drawing a sample, the population must be well-defined. For example, the word “population” can mean a certain group of people, like factory workers, family members, etc., about whom information is needed.

A parameter is a computed value derived from a specific population, such as the mean, the standard deviation, or the standard error of the mean. The value is constant since it applies to the entire population. Statistics, such as mean, standard deviation, and percentage, are values computed from a sample.

There are two primary aims of sampling:

- The evaluation of parameters of the population (mean, proportion, etc.) based on sample statistics.
- To test the population-based hypothesis concerning the sample or samples.

A “sample” is any part of a population that is representative of the whole. Even if a significant number of data samples are gathered from the same group of people, it is still possible that not every single individual is included in the results. The content of samples might vary in terms of size, quality, and sampling procedure; thus, their statistics can also vary. Inferences obtained from a sample pertain solely to the designated population from which the sample or samples were collected. Such findings are applied to the entire population, but generalizations are only true if the samples are sufficiently big and unbiased, i.e., representative of the population from which they are collected. A statistically representative sample will have parameters that are almost identical to those of the total population. There will still be a discrepancy or error of chance, which may be determined from the sample that is representative. The disparities can be diminished but not abolished. A representative sample possesses two primary qualities.

- Precision which means the sample size
- unbiased nature

These two characteristics enable us to accomplish the aforementioned sample objectives. Then, we may determine if observed discrepancies in sample values are attributable to chance or other causes when compared to population characteristics or another sample’s statistics.

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## 3.8 Probability

Probability is the measure of the average relative frequency or likelihood of an event’s occurrence. The primary objective of choosing a fair sample group is to

determine the likelihood (relative frequency) of the occurrence of single or group observations in the normal distribution of any biological variable. We would also like to evaluate the likelihood of sample data points (means and proportions) happening by random chance in order to facilitate the comparison of sample findings to population outcomes. It is therefore possible to rule out random variation and derive conclusions about the population. To do all of this, it is obligatory to have a secure grasp of the probabilities of biological occurrences and events in the population group.

Probability is typically denoted with the sign “ $p$ ”. It varies between zero and one. When “ $p$ ” equals to zero, there is no possibility that an event will occur, or its occurrence is unlikely. If “ $p$ ” equals to one, the probability of an event occurring is 100 percent, i.e., it is unavoidable, such as the death of every living creature.

If the probability of something happening in a population is “ $p$ ” and the probability of it not occurring is given by “ $q$ ”, then

$$q = 1 - p \text{ or } p + q = 1$$

The probability “ $p$ ” or likelihood of a positive event is calculated using the following formula:

$$p = \frac{\text{Number of events occurring}}{\text{Total number of trials}}$$

It is crucial to have a thorough understanding of probability, as it serves as the foundation for all assessments of significance. It is often approximated using the five laws of probability, the normal curve, and the tables.

### 3.8.1 Laws of probability

#### 3.8.1.1 Addition law of probability

The concept of mutually exclusive occurrences refers to situations in which the probability of one event eliminates the possibility of another event or events taking place at the same time. When flipping a coin, receiving head eliminates the possibility of obtaining tail. An occurrence will arise in one of the many possible ways. Thus, occurrences that are mutually exclusive adhere to the addition law of probability. If the count of mutually exclusive occurrences is “ $n$ ” and the individual probability is “ $p^1$ ”, then the overall probability is as follows:

$$P = p^1 + p^2, \dots + p^n = 1$$

Whenever addition law is used, the term “or” is present, e.g., birth of Rh-negative or Rh-positive baby, a medicine will cure or alleviate or have no impact on an ailment.

#### 3.8.1.2 Multiplication law of probability

This law is used when two or more events happen simultaneously, but they must not be related, meaning the events should be independent of one another. When two dice

throws result in an event like 3 and 6 or 6 and 3, the word “and” is used between the occurrences.

If a dice is tossed twice in a row, what is the likelihood of having 3 and 6 or 6 and 3, with the likelihood of obtaining 3 in the first toss being  $1/6$  and 6 in the second being  $1/6$ ?

The likelihood of obtaining 3 in the first toss is  $1/6$  and 6 in the second will be  $1/6$ . So, in first scenario the likelihood of obtaining 3 in the first toss and 6 in the second is  $1/6 \times 1/6 = 1/36$ .

The likelihood of getting 6 on the first toss and 3 on the second would be  $1/6 \times 1/6 = 1/36$  in the second scenario. It demonstrates that order is irrelevant.

The likelihood of obtaining either in two tosses, if the sequences of 3 and 6 or 6 and 3 are ignored, is  $1/36 + 1/36 = 2/36 = 1/18$ .

In certain situations, it is necessary to use both the addition and multiplication principles, and the conjunctions “and” and “or” must be employed. For instance, the Rh factor and birth sex are separate occurrences that can happen to any infant.

### ***3.8.1.3 Binomial law of probability distribution***

Binomial law of probability distribution is established by the terms of the expansion of the binomial expression  $(p + q)^n$ , where “ $n$ ” represents the sample size or the number of events such as births, tosses, or randomly selected individuals for whom the probability is to be calculated, “ $p$ ” represents the probability of ‘success or true or 1,’ “ $q$ ” represents the probability of “failure or false or 0” and  $(p + q)$  is equal to unity.

For instance, when  $n = 2$ , the expansion terms of  $(p + q)^2$  are  $p^2$ ,  $2pq$ , and  $q^2$ . The values of “ $p$ ” and “ $q$ ” are derived from the percentage of population, i.e., the probability of having a baby boy “ $p$ ” or a baby girl “ $q$ ” when just one child is born is determined by watching a huge number of births in the population. It might be 51% for boys ( $p$ ) and 49% for girls ( $q$ ). Substitute  $p = 0.51$  and  $q = (1 - 0.51 = 0.49)$  to determine the probability.

### ***3.8.1.4 Probability (chances) from shape of normal distribution or normal curve***

If the heights of the students in a class follow a normal distribution and the total number of students in the class, let’s say 200, is considered to be unity, then we know that 50 percent of the students in the class are taller than the mean and 50 percent of the students are shorter than the mean. The range, mean  $\pm 1$  SD, includes 68% of children, while the mean  $\pm 2$  SD covers 95%. Therefore, the likelihood of having a height above the mean with  $+2$  standard deviations is 2.5%, and the probability of having a height below the mean with  $-2$  standard deviations is also 2.5%.

It is possible to estimate the likelihood of any observation or number of instances lying above or below the mean at any distance from the mean. Likewise, the whole area beneath the normal curve is assumed to be unity. The proportionate area under any piece of the curve for a normally distributed variable indicates the relative frequency or likelihood of observations between any two places on the horizontal scale.

Similarly, the sampling distribution determines the probability of sample values or results, such as means and proportions, varying by chance from those of the other samples or the population. Theoretically, if a sample is fairly representative of a population, its mean should be the same as that of another representative sample or the population mean. However, this is not what happens in practice. 95% of sample means fall within the range of a mean  $\pm 1.96$  standard error, according to the sampling distribution. 5% probability of the data being higher or lower than this range (0.05 out of 1).

### ***3.8.1.5 Probability of calculated values from tables***

Using the corresponding tables, the probability of computed values occurring by chance for “ $t$ ” and “ $\chi^2$ ” (Chi-square) is obtained. The probabilities or odds of dying or thriving at any age are derived from life tables based on the mortality experience of a large sample of the population that is inclusive of both men and women of all ages. Additionally, modified life table approaches are utilized to determine the probability of survivability at any given time after a therapeutic intervention or procedure.

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## **3.9 Comparing the means of two or more data variables or groups**

By comparing means, the t-test compares the mean of a variable in one group to the mean of the same variable in one, two, or more other sample groups. The null hypothesis is set to zero for the population difference between the two groups. We test this null hypothesis using population sample data. We can perform a one-tailed test (lower than or higher than) or a two-tailed test. For instance, we use one-tailed tests to see if the available data show that the sample mean difference across groups is lower than (or higher than) zero. In many statistical settings, the means of two groups or samples must be compared. The method used to compare means depends on the type of data and how the data is organized ([Pereira and Leslie, 2009](#)). These are the four most common ways to compare means for data that is thought to be normally distributed:

### **3.9.1 Independent samples t-test**

The most commonly used type of t-test is the independent samples t-test, which is also called the unpaired samples t-test. Examining the means of two independent sets of sample data can be useful. If two samples are drawn from the same demographic group for the independent samples t-test, their means may be equivalent. When samples are collected from two diverse populations, the sample mean may differ. In this instance, it is employed to form inferences on the means of two populations and to determine their relationship. For instance, a t-test makes a comparison between the average test results of men and women in response to the question,

“Could these differences have happened by chance?” This independent t-test is applicable in two situations: first, when neither the mean nor the standard deviation of the sample population are known. Second, when two samples are distinct and unrelated to one another.

There are few assumptions in independent samples t-test:

- Independence assumption: two independent and categorical classes are required to express an independent variable. In above example, male and females are two well defined categories and hence, constitutes the independent variables.
- Normality assumption: the dependent variable should have a normal distribution. Additionally, the dependent variable must be evaluated on a continuous measure. In above example, average test results are defined as dependent variables.
- Uniformity of Variance Assumption: the variances should be equivalent for the dependent variables.

### 3.9.2 One sample t-test

It is a statistical test designed to evaluate if the mean calculated from sample data obtained from a single group deviates from a value provided by the researcher. This is an extrinsic value determined for scientific purposes and not obtained from the sample data itself. Typically, this selected value is a previously determined demographic mean, a standard value of concern, or a mean suggested from prior research. The one-sample t-test is like other hypothetical tests in that it checks to see if there is enough information to reject the null hypothesis “ $H_0$ ” in favor of the alternative hypothesis “ $H_1$ .” “The population mean is equal to the given mean value,” is the null hypothesis for a one-sample t-test, while “the population mean is different from the given mean value,” is the alternative hypothesis.

The one sample t-test differs from the vast majority of statistical hypothesis tests in that it neither evaluates the association between variables nor compares two distinct groups. It is a direct measure of comparison between data collected on an individual variable from a specific solitary population and a researcher-determined value.

It is possible to do a one-tailed or two-tailed t-test using the one sample t-test, depending on whether you are looking for a discrepancy from the mean in one direction or in the both directions. An assumption-driven test, the One Sample t-test requires the following:

- Sample data should be independent.
- Sample data is obtained at random as in case of simple random sampling.
- The sample data should have a normal distribution.

### 3.9.3 Paired samples t-test

The paired sample t-test, also known as the dependent sample t-test, is a statistical metric tool for estimating if the mean difference between two sets of observational

data is equal to zero. In this test, each object or entity is measured twice, resulting in two sets of observations. The paired sample t-test is widely used in case-control studies and designs that require repeated measurements. Suppose you wish to determine the efficacy of a program for training students or groups of patients. Using a paired sample t-test, you may examine the differences between the performance of a sample of students or the efficacy of therapy before and after the completion of the process and then look at the significant differences.

The paired sample t-test employs two contradictory research hypotheses, the null hypothesis and the alternative hypotheses, just like most statistical procedures. The null hypothesis states that there is no difference between the mean values of the two paired datasets. According to this perspective, all visible distinctions are the result of random variation. It is possible that the real mean difference between the two samples is not equal to zero, as the alternative hypothesis asserts. The structure of the alternative hypothesis can vary based on the predicted outcome. A two-tailed hypothesis is utilized when the direction of the difference is irrelevant. Alternately, upper- or lower-tailed hypotheses might be used to enhance the validity of the test. Below are the basic definitions of the hypotheses for the paired sample t-test.

- According to the null hypothesis ( $H^0$ ), the actual mean difference ( $\mu^d$ ) equals 0.
- The alternative hypothesis with two tails ( $H^1$ ) indicates that ( $\mu^d$ ) is not equal to 0.
- The alternative hypothesis with an upper tail ( $H^1$ ) indicates that ( $\mu^d$ ) exceeds 0.
- The alternative hypothesis with a lower tail ( $H^1$ ) indicates that ( $\mu^d$ ) is less than 0.

### 3.9.4 ANOVA

Analysis of variance (ANOVA) is an ensemble of statistical theories and their corresponding estimation procedures (such as “variation across or between groups”) that are used to determine the differences between means. ANOVA is governed by the law of total variance, which partitions the observed variance in a variable into components associated with various sources of variation. In its most fundamental form, ANOVA provides a statistical test to evaluate if two or more population means are equivalent, hence expanding the scope of the t-test beyond two means. In other words, when two or more populations have different mean values, ANOVA can be used to determine how significant the difference is.

The assumptions behind the ANOVA test are much the same as underlying for any parametric test.

- ANOVA can only be performed if the subjects in every sample are unrelated. This indicates that subjects in the first group can’t be present in the second group (i.e., independent samples/between-groups).
- The sample sizes of the distinct groups must be equivalent.
- An ANOVA can be performed only if the dependent variable has a normal distribution, in which the middle values are most prevalent and the extreme values are rarest.

- Population variances must be homoscedastic means they should be identical. Homogeneity of variance indicates that the deviation of values (as measured by range or standard deviation) is comparable across populations.

There are different types of ANOVA tests. “One-Way” and “Two-Way” are the most popular types. The distinction between these two categories is controlled by the number of independent variables in an ANOVA test. A one-way ANOVA consists of one categorical independent variable (also called a factor) and one normally distributed continuous dependent variable, for example, testing the effect of a treatment intervention on the likelihood of anxiety in a clinical sample. But a two-way ANOVA consists of two or more categorical independent variables and a single normally distributed continuous dependent variable, for example, employment status is an example of a factorial two-way ANOVA.

### 3.9.5 The Chi-square tests

The Chi-square test is a statistical way to compare two sets of data that doesn't depend on any assumptions or variable distribution. Despite its uniqueness, the Chi-square test has a well-defined distribution that makes it a significant tool for scientific research. It is utilized most frequently when data consist of frequencies, such as the number of responses in two or more population groups. It has three common but crucial uses in biomedical statistics:

- Proportion
- Interrelation
- Integrity of fit

The Chi-square ( $\chi^2$ ) test is a valuable tool for comparing experimentally obtained findings to those theoretically predicted by a given hypothesis. Consequently, the actual difference between the observed and the expected frequencies is captured by the Chi-square statistic. To understand the discrepancy between theory and reality in sampling research, a measurement like this is essential to use. According to the definition, it is as follows:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

where, “O” represents observed frequencies and “E” represents expected frequencies.

Calculating the standard error of difference between two proportions would reveal the relevance of large binomial samples with a size greater than 30. Chi-square test is another extremely helpful test that may be used to determine significance in the same type of data, with two additional benefits.

- To compare the results of two binomial samples, even if the samples are small, fewer than 30.

- To evaluate and compare the frequencies of two multinomial samples such as number of diabetic as well as non-diabetic in different weight groups of individuals.

The Chi-Square test has numerous uses when other parametric tests cannot be utilized. Some applications of Chi-square test are summarized with examples below:

### 3.9.6 Test of independence

This test aids in identifying the relationship between two or more features. Suppose  $N$  observations have been classified according to two features. By implementing this test on the observations (data) provided, we attempt to determine if the features are associated or independent. This relationship could be good, negative, or nonexistent. For instance, we can determine whether there is a correlation between punctuality in class and the percentage of students who pass, and whether paracetamol is successful in reducing fever. The null hypothesis that there is no association is tested to see if the variables under investigation are linked. To put it another way, the two characteristics are distinct from one another.

Once the value of chi square has been calculated, it is compared to the critical threshold value for the given degree of freedom at a certain significance level. As long as the predicted chi-square value is below the critical or threshold table value, we can accept the null hypothesis and say that two characteristics are not connected. If the calculated number is higher than the value in the critical or threshold table, the experiments have shown that the hypothesis is false, so the hypothesis is rejected.

### 3.9.7 Test of goodness of fit

This is the most valuable application of the Chi Square test. This method is utilized mostly for checking the goodness of fit. It seeks to establish whether an observed frequency distribution is distinct from an estimated frequency distribution. When a normal or other sort of ideal frequency curve is fitted to the data, it is important to determine how well this curve corresponds to the observed data.

The following steps are taken to attain the aforementioned objective: In relation to the inquiry, a null and alternative hypothesis are formulated, and a significance level for rejecting the null hypothesis is chosen. A random observational sample is drawn from a suitable statistical sample. Probability is used to calculate theoretical or expected frequencies based on actual observations. Typically, this includes assuming that a particular probability distribution applies to the statistical sample under study. After that, the observed frequencies are compared to the anticipated or theoretical frequencies. If the estimated value of the chi-square is less than the table value at a given level of significance (typically 5%) and for specific degrees of freedom, the fit is regarded as satisfactory. Thus, the discrepancy between the observed and predicted frequencies can be attributed to sampling variations. Alternatively, if the estimated value of the chi-square is greater than the table value, the fit

is regarded as inadequate, i.e., it cannot be attributed to the fluctuations of simple sampling, but rather to the theory's failure to account for the observed facts.

### 3.9.8 Correlation and regression

Correlation and regression are the two most involved strategies for investigating the connection between two quantitative factors. Most of the time, correlation is seen as the question of whether or not there is a link between two things ("x" and "y"). The Spearman's correlation coefficient rho and the Pearson's product-moment correlation coefficient are the two most renowned correlation coefficients.

Conversely, regression gauges the value of the dependent variable in light of the observed value of the autonomous/independent variable, considering the typical measurable statistical connection between at least two factors. There is a lot of equivocalness in figuring out the thought.

### 3.9.9 A look into correlation and regression

For example, when one variable changes, the other also changes. As a result, either directly or indirectly, a shift occurs. It is possible to have uncorrelated variables if one variable does not move in the same direction as another variable. Calculating the degree to which two variables are interconnected is done statistically. It's possible to have both positive and negative correlations. A rise in one variable leads to an equal and opposite increase elsewhere; this is a sign that the two variables are positively linked when they move in the same direction at the same time. As a case study, profit and investment can be cited. The converse is true when two variables move in opposite directions, such as when one rises and the other falls. This scenario is characterized by a negative correlation. Price and demand for a product are excellent examples of this.

When one or more independent variables are altered, Linear Regression can be used to estimate the change in the dependent variable's metric as a result of the change in the average statistically significant association between two or more variables. Linear regression is a powerful and versatile method for predicting the past, present, or future based on past or present occurrences that is used in a number of human activities. An example of this can be found in the following statement: The profits of a company can be calculated using the company's previous results. It is important to note that in a simple linear regression, there are two variables "x" and "y", that are dependent on or influenced by each other. In this context, "y" is referred to as a criterion or dependent variable, whereas "x" is a predictor or independent variable. Below equation provides the regression line for "y" with respect to "x":

$$y = a + bx$$

Here, "a" is a constant and "b" is a regression coefficient.

Correlation and Linear Regression are used when there are two things to measure, like food quantity and the weight; medicine dose and its impact on blood pressure; temperature of the air and its influence on metabolic rate. Another nominal variable, like the name of a species, a study, or a location, keeps the two measurements together.

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### 3.10 Platforms employed for statistical analysis

The majority of the processes employ a Linux system in conjunction with R programming and integrated development environments (IDEs) such as Rstudio, Jupyter Notebook, Vim, etc. For analyzing Next Generation Sequencing (NGS) research, the accessible user interface options are constrained, with Galaxy being by far the only credible choice. DNAnexus as well as artificial intelligence-based RNA-seq (AIR) are two cloud software UIs that are both patented and subscriber based. Massive volumes of genetic data can also be processed using cloud services like Amazon Web Services and Microsoft Azure, however, these are largely controller driven.

#### 3.10.1 Downstream analysis and visualization

Multiple downstream assessments can be done once the study to determine substantially expressed profiles is completed for adding biological context against the described genomes. The two major downstream analysis performed on transcriptome dataset are as follows:

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### 3.11 Gene ontology & pathway analysis

To have a deeper understanding of the biologic systems being investigated in a specific study, one must first comprehend the role and routes by which a gene is expressed in that system. The functionality of genes could be classified using Gene Ontology (GO) categorization methods into 3 groups: cellular components, molecular functions, and biological processes. The GO study, called gene enrichment analysis (GEA) may alternatively be classified into three groups:

#### 3.11.1 Singular enrichment analysis (SEA)

An entry could be a user-defined gene array taken from any high-throughput NGS/microarray study. Depending on the user-defined arbitrary standard, genome inputs are frequently the most important. The genes are classified under the three major areas mentioned before. Fisher's exact test and its variations, such as the EASE score or the Chi-square test, are used in this method to find out if the genotypes are linked

to the functional classification. SEA is performed using techniques like DAVID, GoStat, and Bingo.

### 3.11.2 Gene set enrichment analysis (GSEA)

All genomes within a high-throughput genome study are utilized as inputs for genome set enrichment analysis, guaranteeing that the assessment is independent of any possible bias owing to artificial limits like those employed in SEA. Perhaps genes with minor differentially expressed variations can be evaluated for any further enrichment analysis as a result of this. The ranking of all genome components inside an annotating class is used to determine maximum enrichment scores (MESs). These maximal enrichment scores are evaluated, and the “p-value” is usually found by using Kolmogorov–Smirnov type statistics to compare reported MESs with ones given in a probability sampling or parameterized statistics to measure fold variations across trials. Such an assessment can be done with ErmineJ and FatiScan.

### 3.11.3 Modular enrichment analysis (MEA)

To facilitate phrase linkages, modular enrichment analysis blends SEA type enrichment analysis alongside connectivity search techniques. With this investigation, Kappa estimates of concordance were employed. Genes that do not appear in numerous nearby words are removed from study considering the nature of a Kappa statistic. ADGO, DAVID, as well as GeneCodis, are a few platforms that can execute this type of research. Such gene ontology categorization methods integrate information from a variety of domains, such as KEGG in pathway assessment, Pfam in the case of protein domains, or TRANSFAC for transcriptional regulation.

### 3.11.4 Correlation networks

One important insight gained from the genes is whether or not there is any statistically significant association between the genome list. Even though GEA provides a humongous amount of information regarding genes having similar functions, programmes like GeneMania offer information on gene interactions, resulting in a much more thorough view of the interaction patterns. GeneMania predicts co-expression, co-localization, as well as physical forces amongst genomes, for its users. One such resource is the Biological General Repository for Interaction Datasets (BioGRID), which contains biochemical, genomic, as well as protein-protein interaction data based on known test findings and therefore is updated on a frequent basis. STRING is a library of proteins that interact with each other. It is being used to learn more about how proteins made by genes work together. Weighted gene correlation network analysis (WGCNA), an R-tool that analyses expression profiles either using microarrays or RNA-sequencing to build a correlation network among genes in a specific study, is another option.

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### 3.12 Future prospects and conclusion

In bioinformatics, statisticians have made a key contribution to the development of sophisticated modeling and analytic methods that users may employ to derive relevant biological contents from the vast amounts of multi-platform genomic information. Their in-depth knowledge of the research methods, along with their unpredictability and ambiguity, have particularly qualified them to play a key role in this endeavor. Further research is certainly required in a variety of domains, and new progress is conceivable. Integrative research is another important field. This discipline is just getting underway, and the research establishment is in desperate need of innovative approaches for incorporating data across many systems in order to acquire a much more comprehensive understanding of the cellular biology that underpins it. Such techniques must strike a balance between statistical rigor in constructing links, computing effectiveness in scaling up to massive data environments, and the understandability of outcomes so that our colleagues can understand them effectively.

The statistical evaluation of omics studies is a rapidly expanding area in which agreement on the best techniques might be difficult to come by. Validated experimentally for different testing adjustments that are best suited for relevant outcomes would be created as the study proceeds, and so would improved techniques of data incorporation across diverse resources and systems. Nevertheless, combining clinical and genetic information and fully evaluating hypotheses so that the final results are useful to the overall public remains a significant hurdle in biological information processing. Bioscience and health care have come a long way to where big data is now a normal part of research and treatment. Because of this, statistics has a unique chance to make a big difference in the progress of science by giving other researchers tools that need to have important information in them.

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